Project Summary Report

Executive Summary

Headline:

AlphaEarth embeddings predict Sentinel-1 backscatter strongly for VV and VH, more moderately for polarization-derived targets, and show a weaker but still measurable cross-modal similarity signal.

Top results:

- S1_VV: best result is GBDT (LightGBM) / Embeddings + region + Dynamic World with R2=0.902.

  Ridge / Embeddings only: R2=0.868, RMSE=1.456, MAE=0.938

  Ridge / Embeddings + region + Dynamic World: R2=0.870, RMSE=1.445, MAE=0.934

  GBDT (LightGBM) / Embeddings + region + Dynamic World: R2=0.902, RMSE=1.260, MAE=0.824

  GBDT (LightGBM) / Embeddings only: R2=0.900, RMSE=1.272, MAE=0.834

- S1_VH: best result is GBDT (LightGBM) / Embeddings + region + Dynamic World with R2=0.943.

  Ridge / Embeddings only: R2=0.929, RMSE=1.317, MAE=0.945

  Ridge / Embeddings + region + Dynamic World: R2=0.929, RMSE=1.315, MAE=0.944

  GBDT (LightGBM) / Embeddings + region + Dynamic World: R2=0.943, RMSE=1.175, MAE=0.858

  GBDT (LightGBM) / Embeddings only: R2=0.941, RMSE=1.194, MAE=0.861

Polarization-difference result:

- S1_VV_div_VH: best result is Ridge direct / Embeddings + region + Dynamic World with R2=0.708.

Project Summary Report (cont.)

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Ridge direct / Embeddings only: R2=0.708, RMSE=1.137, MAE=0.801

Structural Ridge baseline / Embeddings only: R2=0.708, RMSE=1.137, MAE=0.801

GBDT (LightGBM) / Embeddings only: R2=0.707, RMSE=1.139, MAE=0.796

Ridge direct / Embeddings + region + Dynamic World: R2=0.708, RMSE=1.137,
MAE=0.801


Structural Ridge baseline / Embeddings + region + Dynamic World: R2=0.708,
RMSE=1.137, MAE=0.801


GBDT (LightGBM) / Embeddings + region + Dynamic World: R2=0.699, RMSE=1.154,
MAE=0.798
```

Direct vs structural follow-up:

- Using the saved LightGBM prediction files, direct VV-VH prediction beats VV_hat - VH_hat on both feature ablations.

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Embeddings only: direct R2=0.707, RMSE=1.139, MAE=0.796; structural R2=0.668,
RMSE=1.212, MAE=0.863.


Embeddings + region + Dynamic World: direct R2=0.699, RMSE=1.154, MAE=0.798;
structural R2=0.677, RMSE=1.196, MAE=0.838.
```

- Derived linear VV/VH benchmark: LightGBM direct ratio (Embeddings only) is strongest at R2=0.584.

Failure-mode readout:

Highest-error S1_VV land-use classes by MAE:

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built: MAE=1.227, bias=-0.333

water: MAE=1.011, bias=0.020
```

Project Summary Report (cont.)

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  shrub_and_scrub: MAE=0.902, bias=-0.054
```

Cross-modal similarity readout:

- At k=10, mean embedding/SAR neighbor overlap is 0.032 versus a random baseline near 0.003.

- Overall embedding-to-SAR distance correlation is modest (Pearson r=0.195), but within-region correlation peaks at 0.689 in california_coast.

Executive takeaway:

- The embeddings carry clear SAR-relevant signal. The strongest wins are on VV and VH, polarization structure is recoverable but weaker, and cross-modal similarity is real but much more convincing within region than across the pooled dataset.

Interpretation and Results Discussion

Interpretation and Results Discussion

- The AlphaEarth embeddings are strongly predictive of absolute Sentinel-1 backscatter. The best held-out results reach R2=0.902 for S1_VV and R2=0.943 for S1_VH, which indicates that the representation captures substantial information about absolute scattering strength.

- The polarization-derived target S1_VV_div_VH, defined here as the dB-space difference VV - VH, is materially harder. Its best held-out performance is R2=0.708, well below the channel-wise results for VV and VH.

- A stricter apples-to-apples strategy comparison built from the saved LightGBM prediction files reaches the same conclusion: for Embeddings only, direct VV-VH prediction scores R2=0.707, RMSE=1.139, MAE=0.796 while VV_hat - VH_hat scores R2=0.668, RMSE=1.212, MAE=0.863. Direct prediction also stays ahead on the second feature ablation.

- The earlier Ridge addendum already favored direct prediction over the structural baseline: direct Ridge reaches R2=0.708, the structural Ridge baseline reaches R2=0.708, and the best LightGBM direct result reaches R2=0.707.

- The similarity between Ridge and LightGBM on S1_VV_div_VH, with an R2 spread of only 0.009 across the evaluated models, indicates that model complexity is not the dominant bottleneck for this target. The more plausible bottleneck is the information content of the features themselves.

- Adding region and Dynamic World context produces only small gains relative to embeddings alone. The total held-out R2 spread across feature sets is 0.002 for S1_VV and 0.002 for S1_VH, which is consistent with those contextual variables being largely redundant with the embedding representation.

Interpretation and Results Discussion (cont.)

- This is a scientifically meaningful result rather than a negative outcome. The present evidence indicates that the AlphaEarth embeddings encode absolute scattering strength better than relative polarization structure, which is a useful characterization of what information the representation preserves.

Plan Alignment

Working Plan Alignment

Source plan: alphaearth-to-sar-cross-model-prediction-eda-working-plan.md

Original planned workflow:

- Phase 1: Data acquisition and preparation

- Phase 2: Regression modeling

- Phase 3: Failure-mode analysis

- Phase 4: Cross-modal similarity

What was actually completed:

- A full-dataset workflow on 2,880 balanced samples instead of the earlier small-subset concept.

- Phase 2 used LightGBM as the primary nonlinear model, with added Ridge-based follow-up baselines for polarization difference and derived ratio analysis.

- Phase 3 and Phase 4 were completed with reportable outputs and summary figures.

- Umbra SAR remains a planned extension rather than a completed dataset in the current pipeline.

Interpretation:

- The project should be described as an updated implementation of the working plan, not a verbatim execution of every proposed modeling choice.

Methods and Data

Methods and Data

Dataset:

- 2,880 co-located samples across four regions and nine Dynamic World labels.

- Targets: S1_VV, S1_VH, and S1_VV_div_VH where the stored polarization target is the dB-space difference VV - VH.

- Feature sets: 64-dim AlphaEarth embeddings alone, or embeddings plus region and Dynamic World context.

Evaluation design:

- 70/30 held-out split stratified by region x land-use label.

- Repeated grouped cross-validation on the training split using spatial blocks inside each region.

- Metrics: R2, RMSE, MAE, Pearson correlation.

Caveat from the working plan context:

- The holdout is stratified random rather than geographically disjoint, so performance should be interpreted as within-region generalization rather than strict transfer to unseen geographies.

Success Criteria Review

Planned Success Criteria vs Observed Results

Predictive capacity:

- The working plan targeted Ridge R2 > 0.5 and boosted-tree R2 > 0.65. Observed best held-out R2 values are VV=0.902, VH=0.943, and VV-VH=0.707.

- The RMSE < 3 dB target is satisfied comfortably for the completed Sentinel-1 targets: VV=1.260, VH=1.175, VV-VH=1.139.

Cross-modal similarity:

- The plan's optimistic k-NN overlap target (>40%) is not met in pooled form; observed mean overlap at k=10 is 0.032.

- The plan's distance-correlation target (>0.3) is approximately met overall with Pearson r=0.195, and is exceeded within some regions.

Failure analysis:

- The plan expected clear land-use-specific error differences. The completed Phase 3 outputs do show materially different MAE by land-use class.

- Strong global evidence of residual spatial clustering was not established; any isolated Moran's I hits should be treated as exploratory.

Best Phase 2 Models by Target

target	feature_set	r2	rmse	mae	pearson_r
S1_VH	Embeddings + region + Dynamic World	0.943	1.175	0.858	0.971
S1_VV	Embeddings + region + Dynamic World	0.902	1.260	0.824	0.951
S1_VV_div_VH	Embeddings only	0.707	1.139	0.796	0.841

These are the best held-out full-dataset models selected for the project summary.

Polarization Difference Comparison

feature_set	model_label	target	training_target	r2	rmse	mae	pearson_r
Embeddings only	Ridge (VH) [Embeddings only]	S1_VH	S1_VH	0.929	1.317	0.945	0.964
Embeddings only	Ridge (VV) [Embeddings only]	S1_VV	S1_VV	0.868	1.456	0.938	0.933
Embeddings only	Ridge (VV-VH) [Embeddings only]	S1_VV_div_VH	S1_VV_div_VH	0.708	1.137	0.801	0.842
Embeddings only	Structural baseline (VV_hat - VH_hat) [Embeddings only]	S1_VV_div_VH	S1_VV+S1_VH	0.708	1.137	0.801	0.842
Embeddings only	LightGBM (VV-VH) [Embeddings only]	S1_VV_div_VH	S1_VV_div_VH	0.707	1.139	0.796	0.841
Embeddings + region + Dynamic World	Ridge (VH) [Embeddings + region + Dynamic World]	S1_VH	S1_VH	0.929	1.315	0.944	0.964
Embeddings + region + Dynamic World	Ridge (VV) [Embeddings + region + Dynamic World]	S1_VV	S1_VV	0.870	1.445	0.934	0.934
Embeddings + region + Dynamic World	Ridge (VV-VH) [Embeddings + region + Dynamic World]	S1_VV_div_VH	S1_VV_div_VH	0.708	1.137	0.801	0.842
Embeddings + region + Dynamic World	Structural baseline (VV_hat - VH_hat) [Embeddings + region + Dynamic World]	S1_VV_div_VH	S1_VV+S1_VH	0.708	1.137	0.801	0.842
Embeddings + region + Dynamic World	LightGBM (VV-VH) [Embeddings + region + Dynamic World]	S1_VV_div_VH	S1_VV_div_VH	0.699	1.154	0.798	0.836

This table compares direct VV-VH prediction, the structural VV_hat - VH_hat baseline, and the reused LightGBM VV-VH result.

Direct vs Structural Follow-up

feature_set	approach	r2	rmse	mae	prediction_file
Embeddings only	Direct LightGBM VV-VH	0.707	1.139	0.796	test_predictions_embedding_only_S1_VV_div_VH.csv
Embeddings only	Structural VV_hat - VH_hat	0.668	1.212	0.863	test_predictions_embedding_only_S1_VV_div_VH_structural_from_vv_minus_vh.csv
Embeddings + region + Dynamic World	Direct LightGBM VV-VH	0.699	1.154	0.798	test_predictions_embedding_plus_context_S1_VV_div_VH.csv
Embeddings + region + Dynamic World	Structural VV_hat - VH_hat	0.677	1.196	0.838	test_predictions_embedding_plus_context_S1_VV_div_VH_structural_from_vv_minus_vh.csv

This follow-up uses the saved LightGBM prediction files to compare direct S1_VV_div_VH prediction against the reconstructed structural baseline VV_hat - VH_hat on the same held-out rows.

Feature Importance Interpretation

Feature Importance Interpretation

The LightGBM feature-importance analysis uses the best held-out LightGBM run for each main target and summarizes permutation importance.

- S1_VV: top features are water (0.113), A27 (0.084), A49 (0.055), A25 (0.041), A17 (0.037).

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Positive importance share by family: embeddings=79.1%, Dynamic World=20.4%,
region=0.5%.
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- S1_VH: top features are water (0.117), A25 (0.105), A27 (0.045), bare (0.030), A49 (0.026).

```
Positive importance share by family: embeddings=70.2%, Dynamic World=29.7%,
region=0.1%.
```

- S1_VV_div_VH: top features are A25 (0.126), A24 (0.047), A63 (0.042), A39 (0.035), A07 (0.033).

```
Positive importance share by family: embeddings=100.0%, Dynamic World=0.0%,
region=0.0%.
```

Interpretation:

- Embedding dimensions dominate the ranked importance lists, while region and Dynamic World features contribute comparatively little mass.

- This reinforces the earlier ablation result that contextual variables add only marginal incremental signal beyond the embedding itself.

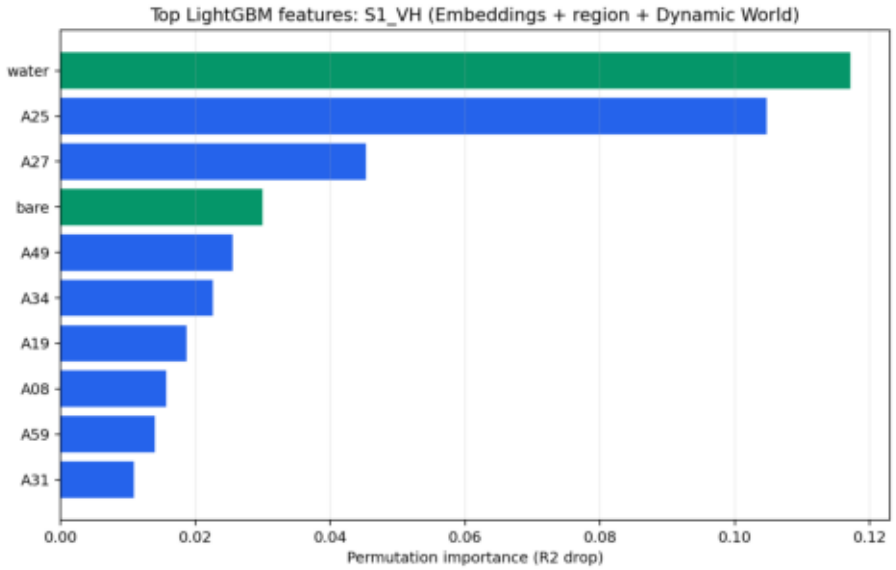
Top LightGBM Features

target	feature_set	feature	feature_family	permutation_mean
S1_VH	Embeddings + region + Dynamic World	water	dynamic_world	0.117
S1_VH	Embeddings + region + Dynamic World	A25	embedding	0.105
S1_VH	Embeddings + region + Dynamic World	A27	embedding	0.045
S1_VH	Embeddings + region + Dynamic World	bare	dynamic_world	0.030
S1_VH	Embeddings + region + Dynamic World	A49	embedding	0.026
S1_VH	Embeddings + region + Dynamic World	A34	embedding	0.023
S1_VH	Embeddings + region + Dynamic World	A19	embedding	0.019
S1_VH	Embeddings + region + Dynamic World	A08	embedding	0.016
S1_VH	Embeddings + region + Dynamic World	A59	embedding	0.014
S1_VH	Embeddings + region + Dynamic World	A31	embedding	0.011
S1_VV	Embeddings + region + Dynamic World	water	dynamic_world	0.113
S1_VV	Embeddings + region + Dynamic World	A27	embedding	0.084
S1_VV	Embeddings + region + Dynamic World	A49	embedding	0.055
S1_VV	Embeddings + region + Dynamic World	A25	embedding	0.041
S1_VV	Embeddings + region + Dynamic World	A17	embedding	0.037
S1_VV	Embeddings + region + Dynamic World	A19	embedding	0.031
S1_VV	Embeddings + region + Dynamic World	A34	embedding	0.031
S1_VV	Embeddings + region + Dynamic World	A39	embedding	0.024
S1_VV	Embeddings + region + Dynamic World	A31	embedding	0.021
S1_VV	Embeddings + region + Dynamic World	bare	dynamic_world	0.019
S1_VV_div_VH	Embeddings only	A25	embedding	0.126
S1_VV_div_VH	Embeddings only	A24	embedding	0.047
S1_VV_div_VH	Embeddings only	A63	embedding	0.042
S1_VV_div_VH	Embeddings only	A39	embedding	0.035
S1_VV_div_VH	Embeddings only	A07	embedding	0.033
S1_VV_div_VH	Embeddings only	A32	embedding	0.030
S1_VV_div_VH	Embeddings only	A30	embedding	0.028
S1_VV_div_VH	Embeddings only	A31	embedding	0.021
S1_VV_div_VH	Embeddings only	A03	embedding	0.019
S1_VV_div_VH	Embeddings only	A62	embedding	0.019

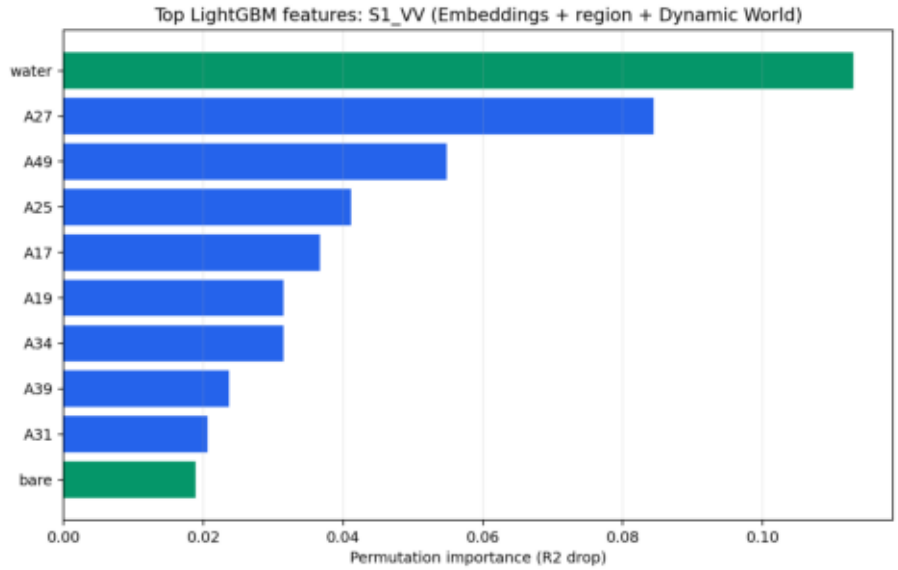
Feature importance is based on the best held-out LightGBM model for each target. Feature family separates embedding dimensions from Dynamic World and region indicators.

Feature Importance Visuals

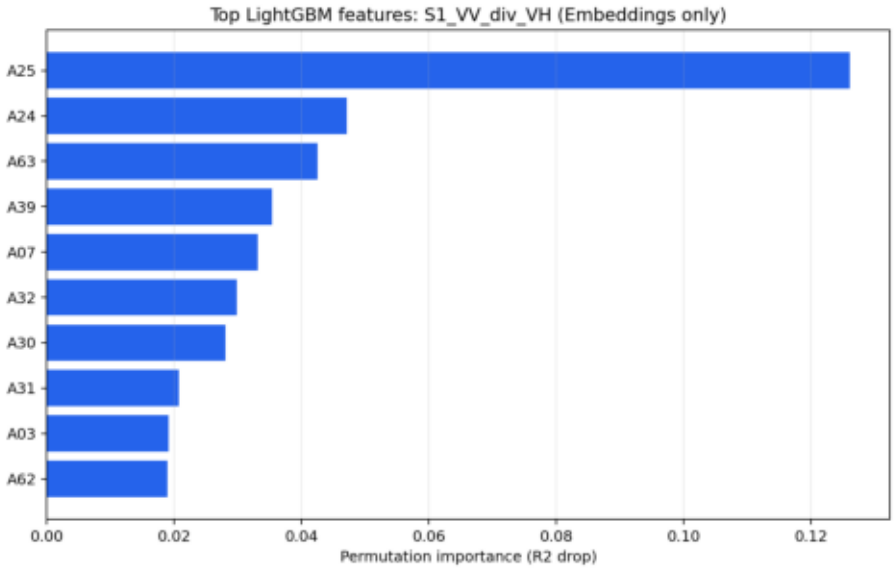
project report feature importance S1 VH



project report feature importance S1 VV



project report feature importance S1 VV div VH



Residual Spatial Pattern Analysis

Residual Spatial Pattern Analysis

- S1_VV: the largest regional error burden appears in sf_bay_urban (MAE=0.981). The strongest positive bias appears in amazon_forest (mean residual=0.028) and the strongest negative bias appears in iowa_ag (mean residual=-0.169).

- S1_VH: the largest regional error burden appears in sf_bay_urban (MAE=0.991). The strongest positive bias appears in iowa_ag (mean residual=0.051) and the strongest negative bias appears in california_coast (mean residual=-0.057).

- S1_VV_div_VH: the largest regional error burden appears in iowa_ag (MAE=0.860). The strongest positive bias appears in amazon_forest (mean residual=0.087) and the strongest negative bias appears in iowa_ag (mean residual=-0.248).

Interpretation:

- The residual patterns are region-dependent rather than completely uniform, but they do not indicate a single catastrophic geographic failure mode across all targets.

- This should be described as modest regional bias structure, not as evidence that the entire model is driven by geography alone.

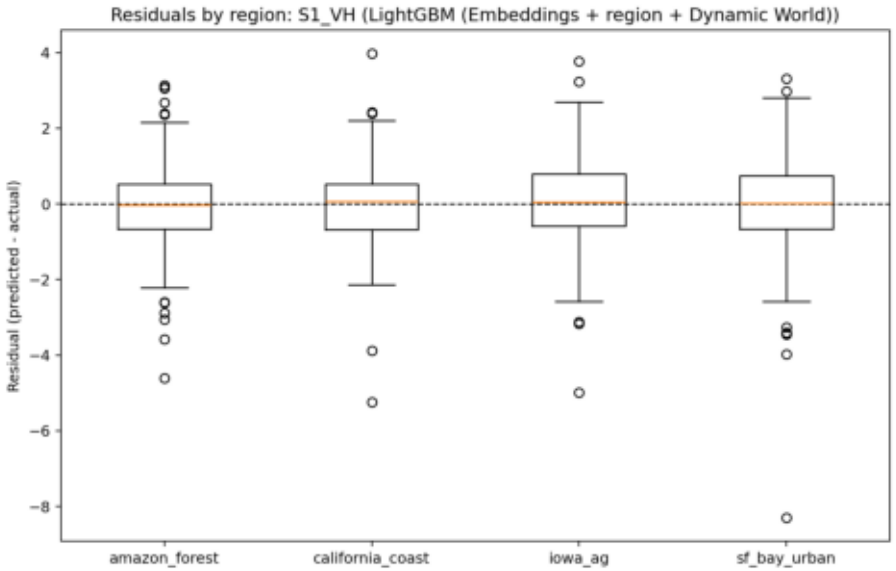
Residual Bias By Region

target	model_label	region	n	mean_residual	median_residual	rmse	mae
S1_VH	LightGBM (Embeddings + region + Dynamic World)	sf_bay_urban	216	-0.002	0.018	1.369	0.991
S1_VH	LightGBM (Embeddings + region + Dynamic World)	iowa_ag	216	0.051	0.043	1.173	0.880
S1_VH	LightGBM (Embeddings + region + Dynamic World)	amazon_forest	216	-0.050	-0.024	1.103	0.812
S1_VH	LightGBM (Embeddings + region + Dynamic World)	california_coast	216	-0.057	0.063	1.027	0.748
S1_VV	LightGBM (Embeddings + region + Dynamic World)	sf_bay_urban	216	0.010	0.012	1.474	0.981
S1_VV	LightGBM (Embeddings + region + Dynamic World)	iowa_ag	216	-0.169	0.011	1.357	0.789
S1_VV	LightGBM (Embeddings + region + Dynamic World)	amazon_forest	216	0.028	-0.032	1.129	0.778
S1_VV	LightGBM (Embeddings + region + Dynamic World)	california_coast	216	-0.014	0.018	1.029	0.750
S1_VV_div_VH	LightGBM (Embeddings only)	iowa_ag	216	-0.248	-0.072	1.366	0.860
S1_VV_div_VH	LightGBM (Embeddings only)	sf_bay_urban	216	0.015	0.014	1.123	0.820
S1_VV_div_VH	LightGBM (Embeddings only)	california_coast	216	0.028	0.115	0.987	0.759
S1_VV_div_VH	LightGBM (Embeddings only)	amazon_forest	216	0.087	0.091	1.042	0.746

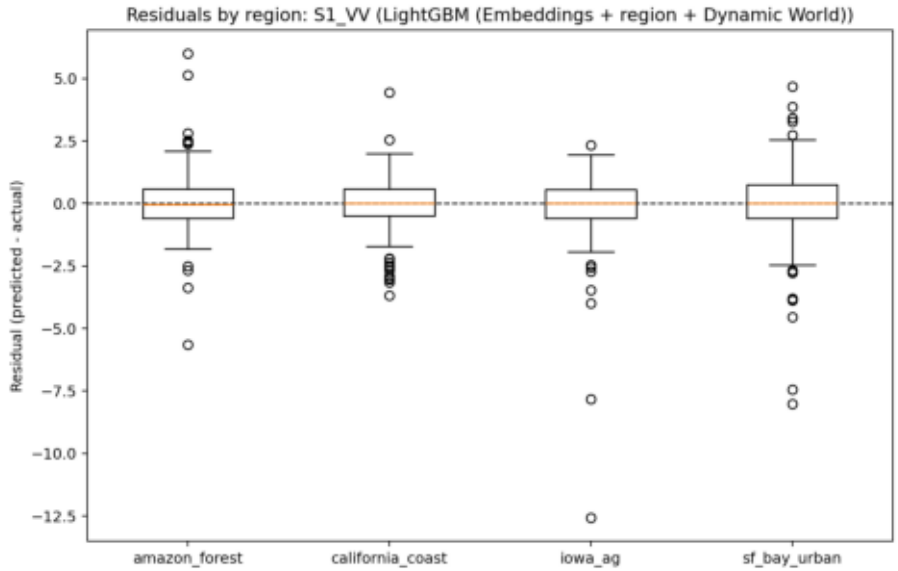
Residuals are computed from the best held-out model per target. Positive mean residual indicates overprediction.

Residual Region Boxplots

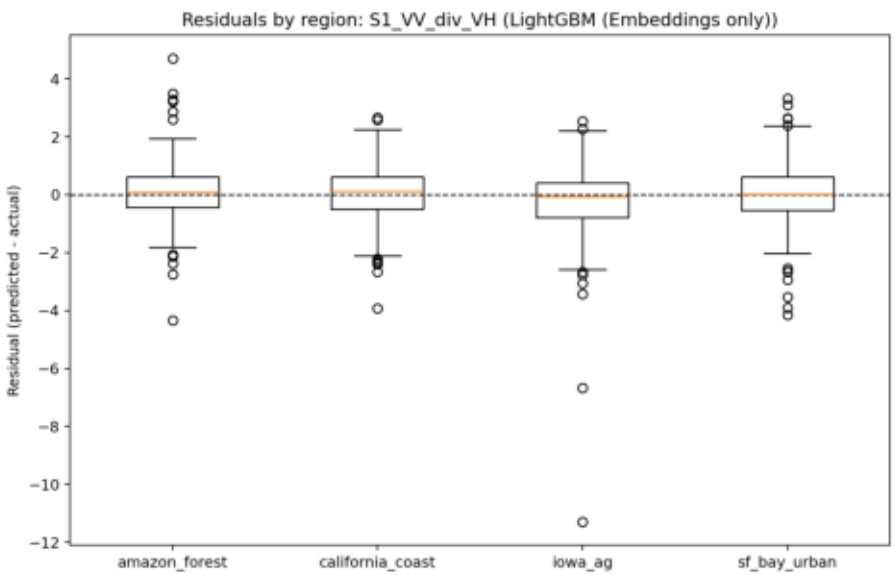
project report residual boxplot S1 VH



project report residual boxplot S1 VV



project report residual boxplot S1 VV div VH



Polarization-difference Failure Analysis

Polarization-difference Failure Analysis

- The failure analysis uses Ridge (VV-VH) [Embeddings + region + Dynamic World], the best held-out predictor for S1_VV_div_VH.

- The hardest region is sf_bay_urban with MAE=0.836 and bias=-0.005.

- The hardest Dynamic World class is water with MAE=1.307 and bias=0.269.

- Error also varies with the true target magnitude. The hardest true-value quintile is (9.014, 18.269] with MAE=1.360.

Interpretation:

- The largest misses for S1_VV_div_VH are concentrated in specific regions and land-cover types rather than being purely random noise.

- Extreme or less typical polarization-difference regimes are harder to recover, which is consistent with the weaker overall performance on this derived target.

S1_VV_div_VH Failure Summary

dw_label_name	n	mae	bias	rmse
water	96	1.307	0.269	1.703
built	96	0.976	-0.174	1.580
bare	96	0.856	0.187	1.148
shrub_and_scrub	96	0.782	0.087	0.977
snow_and_ice	96	0.770	-0.275	1.082
grass	96	0.733	-0.139	0.956
crops	96	0.696	-0.055	0.893
flooded_vegetation	96	0.615	-0.109	0.839
trees	96	0.475	-0.047	0.610

S1_VV_div_VH is the stored dB-space polarization difference VV - VH. The table shows held-out error by land-cover class for the best polarization-difference model.

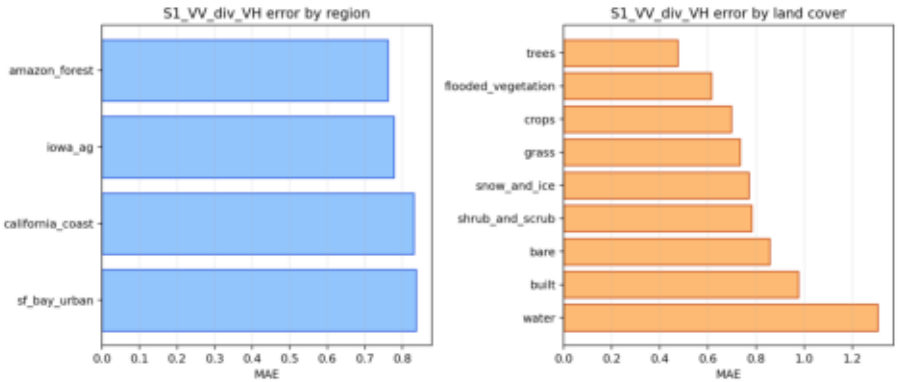
S1_VV_div_VH Outlier Catalog

outlier_rank	region	dw_label_name	latitude	longitude	actual	predicted	residual	abs_residual
1	iowa_ag	built	42.394	-95.867	18.269	8.917	-9.352	9.352
2	iowa_ag	built	41.821	-95.006	16.365	10.485	-5.880	5.880
3	amazon_forest	water	-3.587	-60.026	2.855	7.545	4.689	4.689
4	amazon_forest	water	-3.528	-59.968	3.130	7.277	4.148	4.148
5	amazon_forest	water	-3.065	-60.394	6.047	10.187	4.140	4.140
6	iowa_ag	snow_and_ice	41.912	-95.866	13.064	8.967	-4.097	4.097
7	california_coast	water	35.493	-121.090	16.848	12.784	-4.065	4.065
8	sf_bay_urban	snow_and_ice	37.413	-122.432	13.109	9.054	-4.055	4.055
9	sf_bay_urban	water	37.546	-122.384	6.096	9.695	3.599	3.599
10	sf_bay_urban	bare	37.577	-122.517	5.679	9.274	3.595	3.595
11	amazon_forest	built	-3.073	-60.069	11.403	8.007	-3.397	3.397
12	california_coast	water	35.304	-120.914	9.121	12.468	3.347	3.347
13	california_coast	water	35.483	-121.202	16.736	13.475	-3.261	3.261
14	california_coast	flooded_vegetation	35.475	-120.982	12.140	8.914	-3.226	3.226
15	sf_bay_urban	grass	37.589	-121.981	11.779	8.663	-3.115	3.115

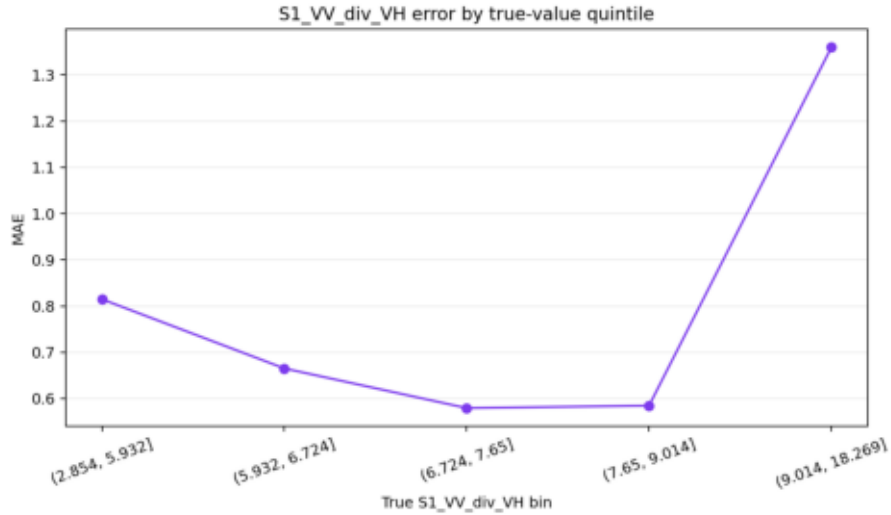
These are the largest held-out absolute errors for the best S1_VV_div_VH model.

Polarization Failure Visuals

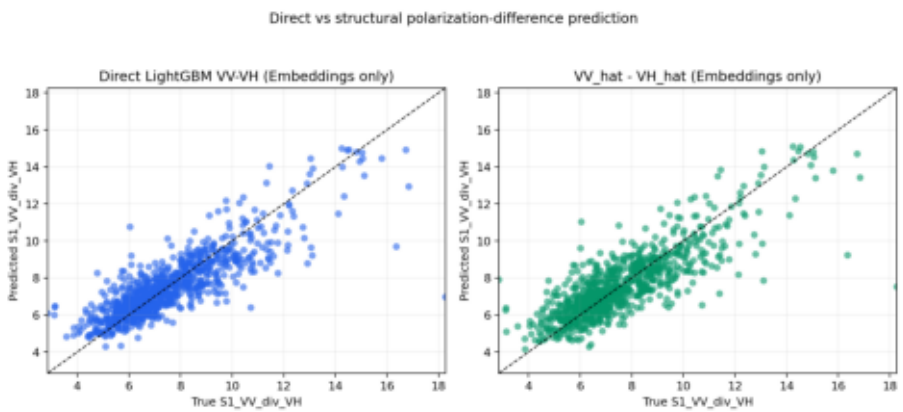
project report polarization failure summary



project report polarization error by value bin



project report polarization direct vs structural



Top S1_VV Failure Modes

dw_label_name	mae	rmse	bias	p90_abs_error	n
built	1.227	2.070	-0.333	2.176	96
water	1.011	1.500	0.020	1.957	96
shrub_and_scrub	0.902	1.366	-0.054	1.742	96
bare	0.837	1.202	0.289	1.697	96
crops	0.772	1.006	0.056	1.594	96

The cumulative report focuses on the highest-error S1_VV land-use classes.

S1_VV Spatial Residual Structure

region	morans_i	p_value	significant_at_0_05	n
amazon_forest	-0.026	0.506	False	216
california_coast	0.092	0.026	True	216
iowa_ag	0.012	0.742	False	216
sf_bay_urban	0.032	0.384	False	216

Significant spatial clustering indicates geographically structured failure modes.

Phase 4 Cross-modal Similarity

k	n_queries	mean_overlap	median_overlap	p10_overlap	p90_overlap
5	2880	0.020	0.000	0.000	0.000
10	2880	0.032	0.000	0.000	0.100
20	2880	0.053	0.000	0.000	0.150

Cross-modal similarity is summarized by k-NN overlap between embedding space and SAR space.

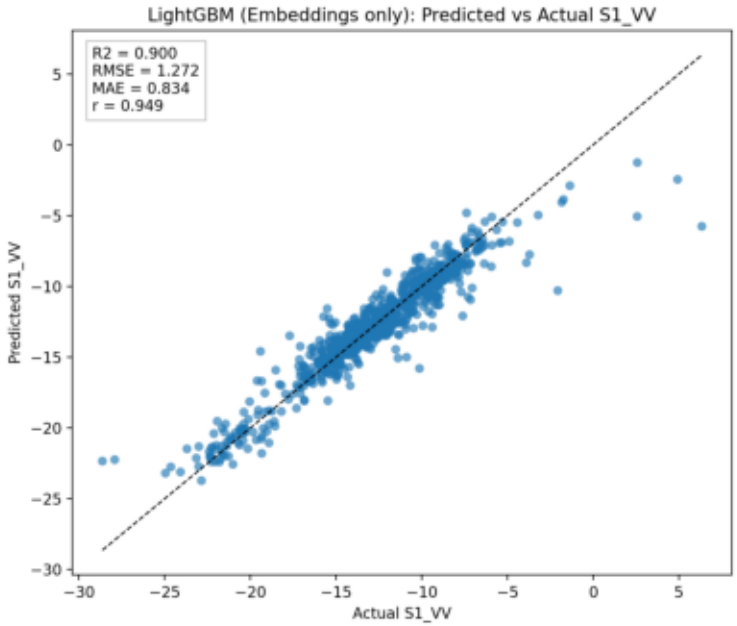
Top Phase 4 Land-use Alignment at k=10

dw_label_name	mean_overlap	median_overlap	p10_overlap	p90_overlap	n_queries
water	0.098	0.100	0.000	0.300	320
snow_and_ice	0.043	0.000	0.000	0.100	320
flooded_vegetation	0.036	0.000	0.000	0.100	320
trees	0.024	0.000	0.000	0.100	320
bare	0.022	0.000	0.000	0.100	320

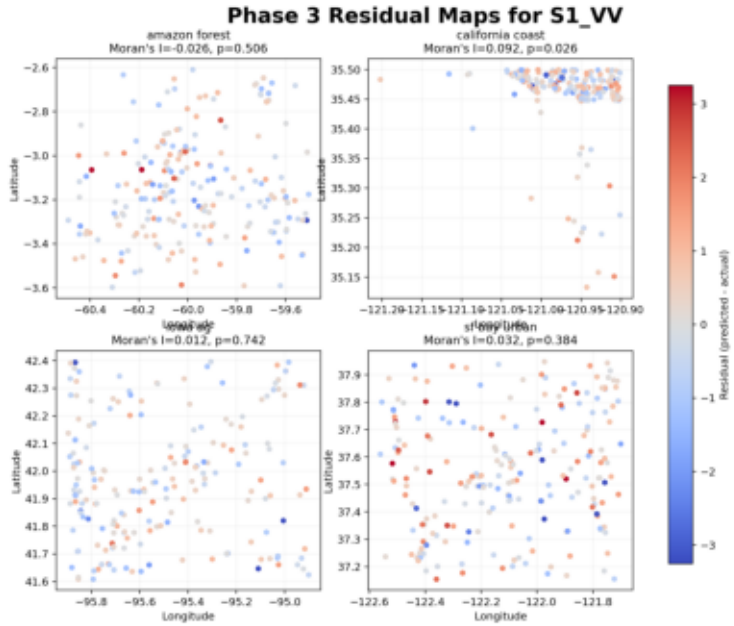
This table highlights which land-use classes show the strongest cross-modal neighbor agreement.

Project Visual Summary

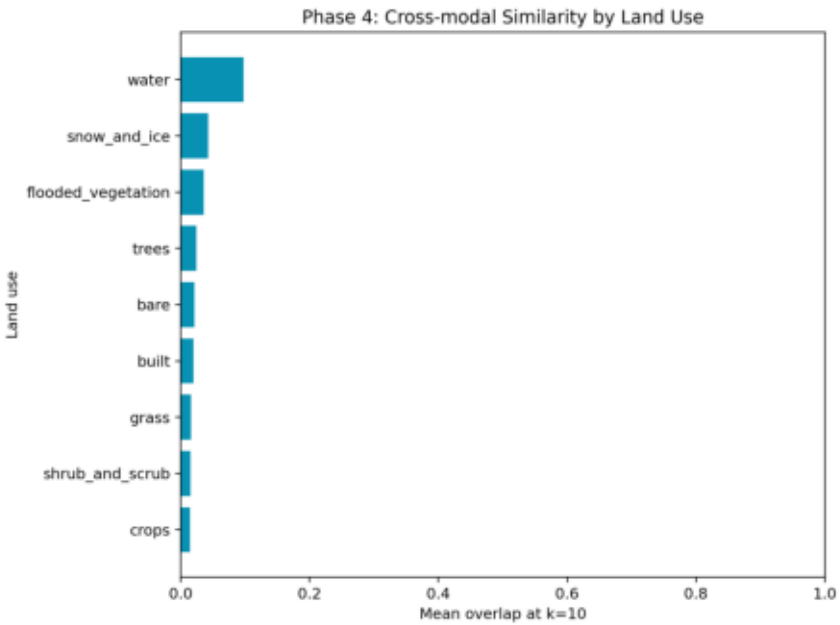
predicted vs actual embedding only S1 VV



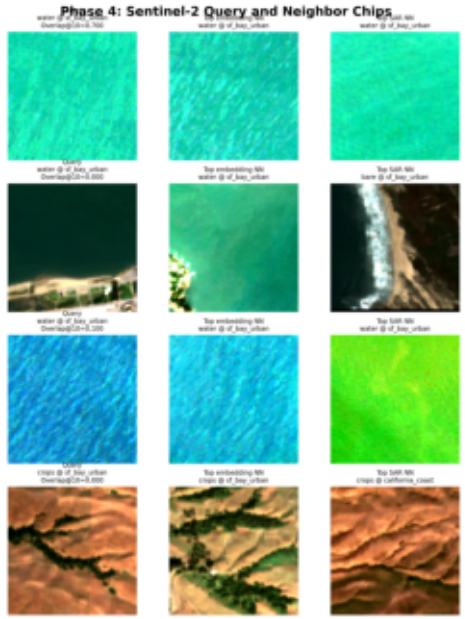
phase3 residual maps S1 VV



phase4 knn overlap by land use k10



phase4 sentinel2 neighbor chips



Next Steps

Recommended Next Steps

- Add Umbra SAR once co-registered scenes are available so the resolution-sensitivity question from the working plan can be answered directly.

- If the next goal is interpretability, prioritize stable Ridge-style analyses for polarization-difference targets and regional subgroup reporting.

- If the next goal is pure predictive performance, focus on stronger spatial holdouts and additional context features rather than more within-region tuning.

- If the next goal is cross-modal representation learning, use the Phase 4 within-region signal as motivation for joint optical/SAR embedding work.